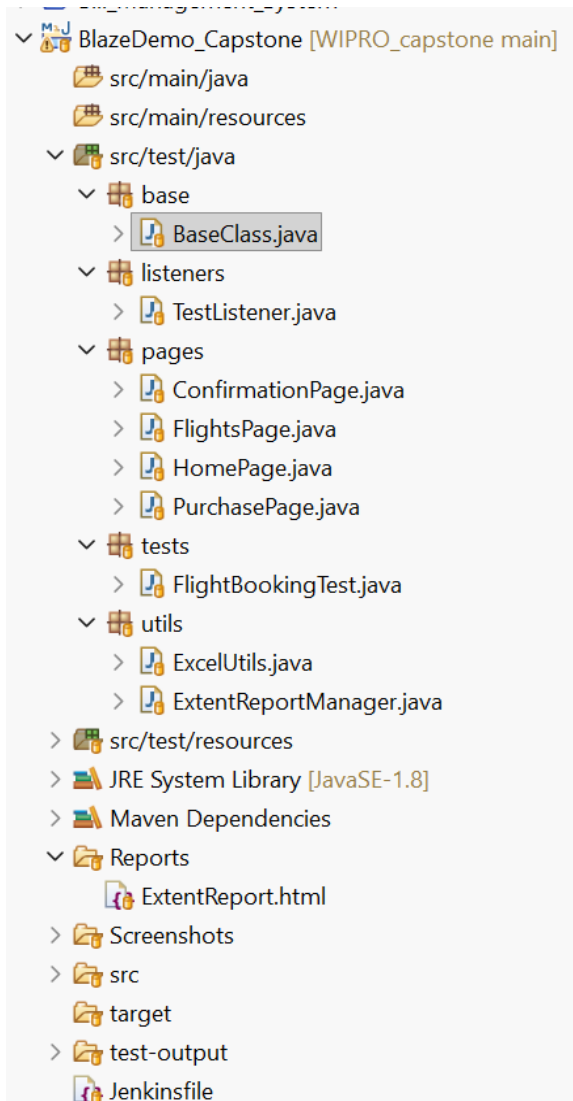


BlazeDemo Capstone Project

Complete Source Code Document



1. BaseClass.java

src/test/java/base/BaseClass.java

```
001 package base;
002
003 import java.io.File;
004 import java.io.IOException;
005 import java.time.Duration;
006
007 import org.apache.commons.io.FileUtils;
008 import org.openqa.selenium.OutputType;
009 import org.openqa.selenium.TakesScreenshot;
010 import org.openqa.selenium.WebDriver;
011 import org.openqa.selenium.chrome.ChromeDriver;
012 import org.openqa.selenium.chrome.ChromeOptions;
013 import org.openqa.selenium.support.ui.WebDriverWait;
014
015 public class BaseClass {
016
017     public WebDriver driver;
018     public WebDriverWait wait;
019
020     public void setup() {
021
022         String executionMode = System.getProperty("executionMode", "local");
023
024         if (executionMode.equalsIgnoreCase("jenkins")) {
025
026             ChromeOptions options = new ChromeOptions();
027
028             options.addArguments("--headless=new");
029             options.addArguments("--disable-gpu");
030             options.addArguments("--window-size=1920,1080");
031             options.addArguments("--remote-allow-origins=*");
032
033             driver = new ChromeDriver(options);
034
035         } else {
036
037             driver = new ChromeDriver();
038         }
039
040         driver.manage().window().maximize();
041
042         driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(10));
043
044         wait = new WebDriverWait(driver, Duration.ofSeconds(15));
045
046         driver.get("https://blazedemo.com/");
047     }
048
049     public void takeScreenshot(String fileName) {
050
051         try {
052
053             File folder = new File("Screenshots");
054
055             if (!folder.exists()) {
056                 folder.mkdir();
```

```
057     }
058
059     File src =
060         ((TakesScreenshot) driver)
061             .getScreenshotAs(OutputType.FILE);
062
063     File dest =
064         new File("Screenshots/" + fileName + ".png");
065
066     FileUtils.copyFile(src, dest);
067
068     } catch (IOException e) {
069
070         e.printStackTrace();
071     }
072 }
073
074 public void tearDown() {
075
076     if (driver != null) {
077         driver.quit();
078     }
079 }
080 }
```

2. TestListener.java

src/test/java/listeners/TestListener.java

```
001 package listeners;
002
003 import org.testng.ITestContext;
004 import org.testng.ITestListener;
005 import org.testng.ITestResult;
006
007 public class TestListener implements ITestListener {
008
009     @Override
010     public void onStart(ITestContext context) {
011         System.out.println("=== Test Execution Started ===");
012     }
013
014     @Override
015     public void onTestStart(ITestResult result) {
016         System.out.println("STARTED : " + result.getName());
017     }
018
019     @Override
020     public void onTestSuccess(ITestResult result) {
021         System.out.println("PASSED : " + result.getName());
022     }
023
024     @Override
025     public void onTestFailure(ITestResult result) {
026         System.out.println("FAILED : " + result.getName());
027         System.out.println(result.getThrowable());
028     }
029
030     @Override
031     public void onTestSkipped(ITestResult result) {
032         System.out.println("SKIPPED : " + result.getName());
033     }
034
035     @Override
036     public void onFinish(ITestContext context) {
037         System.out.println("=== Test Execution Finished ===");
038     }
039 }
```

3. ConfirmationPage.java

src/test/java/pages/ConfirmationPage.java

```
001 package pages;
002
003 import org.openqa.selenium.By;
004 import org.openqa.selenium.WebDriver;
005 import org.testng.Assert;
006
007 public class ConfirmationPage{
008     WebDriver driver;
009
010     By confirmationMessage=By.xpath("//h1[text()='Thank you for your purchase today!']");
011
012     public ConfirmationPage(WebDriver driver){
013         this.driver=driver;
014     }
015
016     public void verifyBookingSuccess(){
017         Assert.assertTrue(driver.findElement(confirmationMessage).isDisplayed(),"Booking confirmation not displayed");
018         System.out.println("Flight booking completed successfully");
019     }
020 }
021 }
```

4. FlightsPage.java

src/test/java/pages/FlightsPage.java

```
001 package pages;
002
003 import java.util.List;
004 import org.openqa.selenium.By;
005 import org.openqa.selenium.WebDriver;
006 import org.openqa.selenium.WebElement;
007 import org.testng.Assert;
008
009 public class FlightsPage {
010     WebDriver driver;
011
012     By flightsTable = By.xpath("//table[@class='table']");
013     By chooseFlights = By.xpath("//input[@value='Choose This Flight']");
014
015     public FlightsPage(WebDriver driver) {
016         this.driver = driver;
017     }
018
019     public void verifyFlightsDisplayed() {
020         Assert.assertTrue(driver.findElement(flightsTable).isDisplayed(), "Flights list not displayed");
021         System.out.println("Flights displayed successfully");
022     }
023
024     public void chooseFlightByIndex(int index) {
025         List<WebElement> flights = driver.findElements(chooseFlights);
026
027         if (index < 1 || index > flights.size()) {
028             throw new RuntimeException("Invalid flight index: " + index);
029         }
030
031         System.out.println("Selecting Flight Index: " + index);
032         flights.get(index - 1).click();
033     }
034 }
```

5. HomePage.java

src/test/java/pages/HomePage.java

```
001 package pages;
002
003 import org.openqa.selenium.By;
004 import org.openqa.selenium.WebDriver;
005 import org.openqa.selenium.support.ui.Select;
006
007 public class HomePage {
008     WebDriver driver;
009
010     By departure = By.name("fromPort");
011     By destination = By.name("toPort");
012     By findFlights = By.xpath("//input[@value='Find Flights']");
013
014     public HomePage(WebDriver driver) {
015         this.driver = driver;
016     }
017
018     public void selectDeparture(String departureCity) {
019         System.out.println("Departure City: " + departureCity);
020         new Select(driver.findElement(departure))
021             .selectByVisibleText(departureCity);
022     }
023
024     public void selectDestination(String destinationCity) {
025         System.out.println("Destination City: " + destinationCity);
026         new Select(driver.findElement(destination))
027             .selectByVisibleText(destinationCity);
028     }
029
030     public void clickFindFlights() {
031         System.out.println("Searching Flights...");
032         driver.findElement(findFlights).click();
033     }
034 }
035 }
```

6. PurchasePage.java

src/test/java/pages/PurchasePage.java

```
001 package pages;
002
003 import org.openqa.selenium.By;
004 import org.openqa.selenium.WebDriver;
005 import org.testng.Assert;
006
007 public class PurchasePage {
008     WebDriver driver;
009
010     By pageHeader = By.xpath("//h2[text()='Your flight from TLV to SFO has been reserved.']").
011     By totalCost = By.xpath("//p[contains(text(),'Total Cost')]");
012     By name = By.id("inputName");
013     By address = By.id("address");
014     By city = By.id("city");
015     By state = By.id("state");
016     By zip = By.id("zipCode");
017     By cardType = By.id("cardType");
018     By cardNumber = By.id("creditCardNumber");
019     By month = By.id("creditCardMonth");
020     By year = By.id("creditCardYear");
021     By nameOnCard = By.id("nameOnCard");
022     By purchaseButton = By.xpath("//input[@value='Purchase Flight']");
023
024     public PurchasePage(WebDriver driver) {
025         this.driver = driver;
026     }
027
028     public void verifyPurchasePage() {
029         Assert.assertTrue(driver.findElement(totalCost).isDisplayed(),
030             "Total cost not displayed");
031         System.out.println("Purchase page displayed successfully");
032     }
033
034     public void enterPersonalDetails(String n, String a, String c, String s, String z) {
035         System.out.println("Passenger Name: " + n);
036         System.out.println("Address: " + a);
037         System.out.println("City: " + c);
038         System.out.println("State: " + s);
039         System.out.println("Zip Code: " + z);
040
041         driver.findElement(name).sendKeys(n);
042         driver.findElement(address).sendKeys(a);
043         driver.findElement(city).sendKeys(c);
044         driver.findElement(state).sendKeys(s);
045         driver.findElement(zip).sendKeys(z);
046     }
047
048     public void enterPaymentDetails(String cn, String m, String y, String noc) {
049         System.out.println("Entering Payment Details");
050
051         driver.findElement(cardNumber).sendKeys(cn);
052
053         driver.findElement(month).clear();
054         driver.findElement(month).sendKeys(m);
055
056         driver.findElement(year).clear();
```



```
057     driver.findElement(year).sendKeys(y);
058
059     driver.findElement(nameOnCard).sendKeys(noc);
060 }
061
062 public void clickPurchaseFlight() {
063     System.out.println("Purchasing Flight...");
064     driver.findElement(purchaseButton).click();
065 }
066 }
```

7. FlightBookingTest.java

src/test/java/tests/FlightBookingTest.java

```
001 package tests;
002
003 import base.BaseClass;
004 import com.aventstack.extentreports.ExtentReports;
005 import com.aventstack.extentreports.ExtentTest;
006 import listeners.TestListener;
007 import org.testng.annotations.Listeners;
008 import org.testng.annotations.AfterMethod;
009 import org.testng.annotations.BeforeMethod;
010 import org.testng.annotations.DataProvider;
011 import org.testng.annotations.Test;
012 import pages.ConfirmationPage;
013 import pages.FlightsPage;
014 import pages.HomePage;
015 import pages.PurchasePage;
016 import utils.ExcelUtils;
017 import utils.ExtentReportManager;
018
019 @Listeners(TestListener.class)
020 public class FlightBookingTest extends BaseClass {
021
022     HomePage homePage;
023     FlightsPage flightsPage;
024     PurchasePage purchasePage;
025     ConfirmationPage confirmationPage;
026
027     ExtentReports extent;
028     ExtentTest test;
029
030     @BeforeMethod
031     public void launchApp() {
032
033         setup();
034
035         driver.get("https://blazedemo.com/");
036
037         homePage = new HomePage(driver);
038         flightsPage = new FlightsPage(driver);
039         purchasePage = new PurchasePage(driver);
040         confirmationPage = new ConfirmationPage(driver);
041
042         extent = ExtentReportManager.getReportInstance();
043     }
044
045     @DataProvider(name = "excelData")
046     public Object[][] getExcelData() {
047
048         ExcelUtils.setExcelFile(
049             "src/test/resources/testdata.xlsx",
050             "FlightData");
051
052         int rows = ExcelUtils.getRowCount();
053
054         Object[][] data = new Object[rows][12];
055
056         for (int i = 1; i <= rows; i++) {
```

```

057         for (int j = 0; j < 12; j++) {
058             data[i - 1][j] = ExcelUtils.getCellData(i, j);
059         }
060     }
061 }
062 }
063
064     return data;
065 }
066
067 @Test(dataProvider = "excelData")
068 public void verifyEndToEndFlightBooking(String departure,String destination,
069     String name,String address,String city, String state,
070     String zip, String cardNumber, String month, String year, String nameOnCard,
071     String flightIndexValue) {
072
073     test = extent.createTest(
074         "Flight Booking Test - " + departure + " to " + destination);
075
076     System.out.println("Running test for: " + departure + " to " + destination);
077
078     int flightIndex = Integer.parseInt(flightIndexValue);
079
080     homePage.selectDeparture(departure);
081
082     homePage.selectDestination(destination);
083
084     homePage.clickFindFlights();
085
086     test.pass("Flight Search Completed");
087
088     flightsPage.verifyFlightsDisplayed();
089
090     test.pass("Flights List Displayed");
091
092     flightsPage.chooseFlightByIndex(flightIndex);
093
094     test.pass("Flight Selected");
095
096     purchasePage.verifyPurchasePage();
097
098     test.pass("Purchase Page Displayed");
099
100     purchasePage.enterPersonalDetails(name,address,city,state,zip);
101
102     test.pass("Personal Details Entered");
103
104     purchasePage.enterPaymentDetails(cardNumber,month, year,nameOnCard);
105
106     test.pass("Payment Details Entered");
107
108     purchasePage.clickPurchaseFlight();
109
110     test.pass("Purchase Flight Button Clicked");
111
112     confirmationPage.verifyBookingSuccess();
113
114     test.pass("Booking Completed Successfully");
115
116     takeScreenshot("BookingConfirmed_" + departure + "_to_" + destination);
117 }

```

```
118
119     @AfterMethod
120     public void closeApp() {
121
122         extent.flush();
123
124         tearDown();
125     }
126 }
```

8. ExcelUtils.java

src/test/java/utis/ExcelUtils.java

```
001 package utis;
002
003 import java.io.FileInputStream;
004 import org.apache.poi.ss.usermodel.*;
005 import org.apache.poi.xssf.usermodel.XSSFWorkbook;
006
007 public class ExcelUtils {
008     static Workbook workbook;
009     static Sheet sheet;
010
011     public static void setExcelFile(String path, String sheetName) {
012         try {
013             FileInputStream fis = new FileInputStream(path);
014             workbook = new XSSFWorkbook(fis);
015             sheet = workbook.getSheet(sheetName);
016         } catch (Exception e) {
017             System.out.println("Excel file error: " + e.getMessage());
018         }
019     }
020
021     public static String getCellData(int row, int col) {
022         String data = "";
023         try {
024             Cell cell = sheet.getRow(row).getCell(col);
025             DataFormatter formatter = new DataFormatter();
026             data = formatter.formatCellValue(cell);
027         } catch (Exception e) {
028             System.out.println("Cell data error: " + e.getMessage());
029         }
030         return data;
031     }
032
033     public static int getRowCount() {
034         return sheet.getLastRowNum();
035     }
036 }
037 }
```

9. ExtentReportManager.java

src/test/java/utils/ExtentReportManager.java

```
001 package utils;
002
003 import com.aventstack.extentreports.ExtentReports;
004 import com.aventstack.extentreports.reporter.ExtentSparkReporter;
005
006 public class ExtentReportManager {
007
008     private static ExtentReports extent;
009
010     public static ExtentReports getReportInstance() {
011
012         if (extent == null) {
013
014             ExtentSparkReporter spark =
015                 new ExtentSparkReporter("Reports/ExtentReport.html");
016
017             extent = new ExtentReports();
018
019             extent.attachReporter(spark);
020         }
021
022         return extent;
023     }
024 }
```

10. pom.xml

pom.xml

```
001 <project xmlns="http://maven.apache.org/POM/4.0.0"
002         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
003         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
004
005     <modelVersion>4.0.0</modelVersion>
006
007     <groupId>com.blazedemo</groupId>
008     <artifactId>BlazeDemo_Capstone</artifactId>
009     <version>1.0</version>
010
011     <properties>
012         <maven.compiler.source>1.8</maven.compiler.source>
013         <maven.compiler.target>1.8</maven.compiler.target>
014         <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
015     </properties>
016
017     <dependencies>
018
019         <!-- Selenium -->
020         <dependency>
021             <groupId>org.seleniumhq.selenium</groupId>
022             <artifactId>selenium-java</artifactId>
023             <version>4.21.0</version>
024         </dependency>
025
026         <!-- TestNG -->
027         <dependency>
028             <groupId>org.testng</groupId>
029             <artifactId>testng</artifactId>
030             <version>7.10.2</version>
031         </dependency>
032
033         <!-- WebDriverManager -->
034         <dependency>
035             <groupId>io.github.bonigarcia</groupId>
036             <artifactId>webdrivermanager</artifactId>
037             <version>5.8.0</version>
038         </dependency>
039
040         <!-- Apache POI -->
041         <dependency>
042             <groupId>org.apache.poi</groupId>
043             <artifactId>poi</artifactId>
044             <version>5.2.5</version>
045         </dependency>
046
047         <dependency>
048             <groupId>org.apache.poi</groupId>
049             <artifactId>poi-ooxml</artifactId>
050             <version>5.2.5</version>
051         </dependency>
052
053         <!-- Extent Reports -->
054         <dependency>
055             <groupId>com.aventstack</groupId>
056             <artifactId>extentreports</artifactId>
```

```
057         <version>5.1.1</version>
058     </dependency>
059
060     <!-- Commons IO -->
061     <dependency>
062         <groupId>commons-io</groupId>
063         <artifactId>commons-io</artifactId>
064         <version>2.16.1</version>
065     </dependency>
066
067     <!-- Log4j -->
068     <dependency>
069         <groupId>org.apache.logging.log4j</groupId>
070         <artifactId>log4j-api</artifactId>
071         <version>2.23.1</version>
072     </dependency>
073
074     <dependency>
075         <groupId>org.apache.logging.log4j</groupId>
076         <artifactId>log4j-core</artifactId>
077         <version>2.23.1</version>
078     </dependency>
079
080 </dependencies>
081
082 <build>
083     <plugins>
084
085         <plugin>
086             <groupId>org.apache.maven.plugins</groupId>
087             <artifactId>maven-compiler-plugin</artifactId>
088             <version>3.11.0</version>
089             <configuration>
090                 <source>1.8</source>
091                 <target>1.8</target>
092             </configuration>
093         </plugin>
094
095         <plugin>
096             <groupId>org.apache.maven.plugins</groupId>
097             <artifactId>maven-surefire-plugin</artifactId>
098             <version>3.2.5</version>
099             <configuration>
100                 <suiteXmlFiles>
101                     <suiteXmlFile>testng.xml</suiteXmlFile>
102                 </suiteXmlFiles>
103             </configuration>
104         </plugin>
105
106     </plugins>
107 </build>
108
109 </project>
```


11. testng.xml

testng.xml

```
001 <!DOCTYPE suite SYSTEM "https://testng.org/testng-1.0.dtd">
002 <suite name="BlazeDemo Capstone Suite">
003 <test name="Flight Booking Test">
004 <classes>
005 <class name="tests.FlightBookingTest"/>
006 </classes>
007 </test>
008 </suite>
```

12. Jenkinsfile

Jenkinsfile

```
001 pipeline {
002     agent any
003
004     tools {
005         jdk 'JDK'
006         maven 'Maven'
007     }
008
009     stages {
010
011         stage('Checkout Code') {
012             steps {
013                 git branch: 'main',
014                 url: 'https://github.com/NishantKumar7007/WIPRO_Capstone.git'
015             }
016         }
017
018         stage('Clean Project') {
019             steps {
020                 bat 'mvn clean'
021             }
022         }
023
024         stage('Compile Project') {
025             steps {
026                 bat 'mvn compile'
027             }
028         }
029
030         stage('Execute TestNG Tests') {
031             steps {
032                 bat 'mvn test -DexecutionMode=jenkins'
033             }
034         }
035
036         stage('Archive Reports') {
037             steps {
038                 archiveArtifacts artifacts: 'test-output/**/*.*', allowEmptyArchive: true
039                 archiveArtifacts artifacts: 'Screenshots/**/*.*', allowEmptyArchive: true
040                 archiveArtifacts artifacts: 'Reports/**/*.*', allowEmptyArchive: true
041             }
042         }
043     }
044
045     post {
046         success {
047             echo 'BlazeDemo Automation Build Successful'
048         }
049
050         failure {
051             echo 'BlazeDemo Automation Build Failed'
052         }
053
054         always {
055             echo 'Pipeline Execution Completed'
056         }
057     }
058 }
```